

Reviewer Pack — Minimal Rank of Geometrically Defined Lattices vs. Quantum C...

Entry 2e386440ed69 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Geometrically Defined Lattices vs. Quantum Circuit Depth

Entry ID: 2e386440ed69 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 20:09:50 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Lattice Theory) **Field B** (complexity object): Complexity Theory: Quantum Circuit Complexity

Statement:

[‘For a given n -vertex graph, the minimal rank of its geometrically defined lattice is $\Theta(n^{\{1.5\}})$. This implies that any quantum circuit with depth D can be represented as a product of at most $O(D^2)$ gates.’, ‘For all instances of graphs with vertex set V , if $|V| \leq 40$ and the geometrically defined lattice $L(G)$ associated with G has rank $r(L(G)) = \Theta(n^{\{1.5\}})$, then there exists a quantum circuit of depth D such that $D \leq O(r(L(G)))^2$.’, ‘Conversely, for any quantum circuit of depth D , there exists an n -vertex graph G with $|V| \leq 40$ and geometrically defined lattice $L(G)$ such that $r(L(G)) = \Theta(n^{\{1.5\}})$ and the circuit can be represented as a product of at most $O(D^2)$ gates.’]

Rationale (proposer’s reasoning):

[‘Lattice theory provides a rich structure for understanding the combinatorial properties of graphs, and geometric lattices have been studied in algebraic combinatorics for their elegant mathematical properties. Quantum circuit depth is a fundamental measure in quantum complexity theory, and it has not been directly linked to lattice theory before.’, ‘This bridge could expose new connections between classical and quantum computing by demonstrating how the geometric structure of graphs translates into properties of quantum circuits.’, ‘It also suggests that problems related to the geometry of graphs might be amenable to quantum algorithms.’]

Taxonomy category: LATTICE_TO_QUANTUM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): fd1572f2c2a04301

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all n -vertex graphs with $|V| \leq 40$, the geometrically defined lattice's rank $r(L(G))$ is within a factor of 1.5 of $n^{\{1.5\}}$ and the corresponding quantum circuit depth D satisfies $D \leq O(r(L(G)))^2$. The conjecture is falsified if there exists an instance where either $r(L(G)) \neq \Theta(n^{\{1.5\}})$ or $D > O(r(L(G)))^2$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL PROOFS	SAFE	0.90	UNCERTAIN	SAFE
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal rank" [geometrically defined] lattice AND "quantum circuit depth" - "algebraic combinatorics" [lattice theory] AND "complexity theory" [quantum circuit complexity] - "graph vertex set" [vertex set ≤ 40] AND [quantum circuit depth] = $O([r(L(G))]^2)$

Top relevant hits considered: - [http://arxiv.org/abs/1710.07343v1] Fractional Minimal Rank - [http://arxiv.org/abs/2208.01392v2] The Analytic Minimal Rank Sard Conjecture - [http://arxiv.org/abs/2406.15089v2] Sentences over Random Groups II: Sentences of Minimal Rank - [http://arxiv.org/abs/1611.00827v2] Geometric complexity theory and matrix powering - [http://arxiv.org/abs/2504.04416v1] Meta-Mathematics of Computational Complexity Theory - [http://arxiv.org/abs/cs/9811005v1] Writing and Editing Complexity Theory: Tales and Tools - [http://arxiv.org/abs/2605.05444v1] Searches for Binary Mergers with Sub-solar Mass Components in Data from the First Part of LIGO-Virgo-KAGRA's Fourth Ob - [http://arxiv.org/abs/2512.17990v2] Constraints on gravitational waves from the 2024 Vela pulsar glitch

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(matrix):
        rows, cols = len(matrix), len(matrix[0])
        rank = 0

        for i in range(cols):
            max_row = rank
            for j in range(rank, rows):
```

```

        if abs(matrix[j][i]) > abs(matrix[max_row][i]):
            max_row = j

    if matrix[max_row][i] == 0:
        continue

    matrix[rank], matrix[max_row] = matrix[max_row], matrix[rank]

    for j in range(rows):
        if j != rank:
            factor = -matrix[j][i] / matrix[rank][i]
            for k in range(cols):
                matrix[j][k] += factor * matrix[rank][k]

    rank += 1

    return rank

def compute_lattice_rank(graph):
    n = len(graph)
    augmented_matrix = [[0] * (n + 1) for _ in range(n)]

    for i in range(n):
        for j in range(n):
            if graph[i][j]:
                augmented_matrix[i][j] = 1
                augmented_matrix[i][-1] += 1

    return gaussian_elimination(augmented_matrix)

def generate_random_graph(n):
    graph = [[0] * n for _ in range(n)]

    for i in range(n):
        for j in range(i + 1, n):
            if random.random() < 0.5:
                graph[i][j] = 1
                graph[j][i] = 1

    return graph

def compute_quantum_circuit_depth(graph):
    # Placeholder for actual quantum circuit depth computation
    # This is a stub and should be replaced with actual logic
    n = len(graph)
    return random.randint(1, n**2) # Random depth between 1 and n^2

n = random.choice([5, 10, 15, 20, 30, 40])
graph = generate_random_graph(n)
lattice_rank = compute_lattice_rank(graph)
quantum_circuit_depth = compute_quantum_circuit_depth(graph)

metric_name = "Quantum Circuit Depth vs. Lattice Rank"
metric_value = quantum_circuit_depth
instances_tested = 1

```

```

conjecture_holds = False
counterexample = ""

if lattice_rank == 0:
    counterexample = "Lattice rank is zero, which is not  $O(n^{1.5})$ "
elif abs(lattice_rank - n**1.5) / (n**1.5) <= 0.75 and quantum_circuit_depth <= lattice_rank**1.5:
    conjecture_holds = True
else:
    counterexample = f"Counterexample: lattice rank {lattice_rank}, depth {quantum_circuit_depth}"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys

    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='{result['counterexample']}'\n first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32f7098e.py", line 109, in <module>

```

    trial_result = run_trial(seed)
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32f7098e.py", line 77, in run_trial

```

    lattice_rank = compute_lattice_rank(graph)
    ~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32f7098e.py", line 56, in compute_latti
    return gaussian_elimination(augmented_matrix)
    ~~~~~^~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32f7098e.py", line 31, in gaussian_elim
    if matrix[max_row][i] == 0:
    ~~~~~^~~~~~

```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:
 skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:
 The test crashed before producing data, which means it did not complete its execution to verify the conjecture. | next: Re-run the test and ensure it completes without crashing to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16968
2	preregistration	ollama_remote	glm4:latest	0	0	10427
3	novelty	ollama_remote	glm4:latest	0	0	8830
4	novelty	ollama_remote	glm4:latest	0	0	9560
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13289
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10550
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11187
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12857
9	judge	ollama_remote	glm4:latest	0	0	12686

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 106354 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/2e386440ed69.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/2e386440ed69.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/2e386440ed69.tar.gz (if generated)